

QUESTION BANK

UNIT-1

1. Explain the contribution of Aryabhata to Indian mathematics.
2. What are the four parts of Aryabhatiya? Briefly describe each.
3. Explain the mathematical achievements of Srinivasa Ramanujan in number theory.
4. Discuss the role of Swami Bharati Krishna Tirthaji Maharaj in rediscovering Vedic Mathematics.
5. Explain the sutra *Ekadhikena Purvena* with two step-by-step examples.
6. Describe the use of *Nikhilam Navatashcaramam Dashatah* in subtraction with an example.

UNIT-2

1. Find Complement of 103 with respect to 100.
2. Find complement of 467 with respect to 1000.
3. Multiply 98×97 by using complement.
4. Find the digital root of $(256 \times 18) + 37$.
5. Construct the table of 9 using the Nikhilam Sutra.

Unit-3

❖ Using *Ekanyunena Purvena* (एकान्यूननेन पूर्वेण)

- 1) 999×738
- 2) 9999×4567
- 3) 9999×8321
- 4) 999×901

❖ Multiplication with अन्त्ययोर्दसाकेनप
 84×86

❖ *Urdhva-Tiryagbhyam* (ऊर्ध्व-तिर्यग्भ्याम्)

$$23 \times 14$$

❖ **Ekadhikena Purvena (एकाधिकेन पूर्वेण)**

1) $(95)^2$

❖ **यावदूनम् (Yāvadūnam)** method of squaring numbers.

1) 98^2

❖ **Square root of a perfect square**

1) 20736

UNIT-4

1. Find addition of polynomial $5x^5 + 6x^3 + 8x + 100$ and $10x^2 + 11x + 11$.
2. Subtract $x - 5$ from $4x^4 - 5x^3 + 6x^2 + 8$.
3. Use the Urdhva Triyabhyam sutra to multiply $4x^2 + 2x + 3$ and $3x^2 + 3x + 2$.
4. Use the Urdhva Triyabhyam sutra to multiply $2x^2 + 3x + 4$ and $5x^2 + 6x + 7$.
5. Use the Urdhva Triyabhyam sutra to multiply $x^2 + 5$ and $x^2 - 11x + 6$.
6. Use the Urdhva Triyabhyam sutra to multiply $(2x + 1)$, $(3x + 3)$ and $(x - 5)$.
7. Use the concept of Paravartya to find $\frac{2x^2+4x+7}{x+2}$.
8. Use the concept of Paravartya to find remainder and quotient of $\frac{4x^2-4}{x-1}$.
9. Use the concept of Paravartya to find remainder and quotient of $\frac{x^2+1}{x-1}$.
10. Use the concept of Paravartya to find remainder and quotient of $\frac{x^3-1}{x+1}$.
11. Factorize quartile $4x^2 + 12x + 5$ use the concept of Anurupyena and Adyamanyantyananteya.

UNIT-5

1. Find HCF(12,15,21) and LCM(12,15,21).

2. Find HCF(50,70,85) and LCM(50,70,85).
3. Find HCF(10,30,50,100) and LCM(10,30,50,100).
4. Find HCF(8,20,32) and LCM(8,20,32).
5. Find HCF(10,15,100) and LCM(10,15,100).
6. Find HCF(3,9,18,27) and LCM(3,9,18,27).
7. Find LCM(6,12,18) and HCF(6,12,18).
8. Solve following bivariate linear equations.

1) $3x + 5y = 19$

2) $2x + 3y = 12$

9. Solve following bivariate linear equations.

1) $x + y = 8$

2) $2x - 3y = 1$

10. Solve equation $\frac{3x+4}{6x+7} = \frac{5x+6}{2x+3}$.

11. Solve equation $\frac{1}{x+7} + \frac{1}{x+9} = \frac{1}{x+6} + \frac{1}{x+10}$.

12. Solve equation $\frac{1}{x-7} + \frac{1}{x+9} = \frac{1}{x+11} + \frac{1}{x-9}$.